Algorithmics	Student information	Date	Number of session
	UO: 300737	17/02/2025	2
	Surname: Hoyos		
	Name: Ignacio		

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Activity 1: Measuring execution times

TABLE1 (times in milliseconds WITHOUT OPTIMIZATION)

n	tLoop1	tLoop2	tLoop3	tLoop4
100	0,016	0,41	1,56	1,37
200	0,033	1,54	6,35	9,68
400	0,0701	7,63	27,7	73,02
800	0,166	33,87	123,85	672,62
1600	0,3674	129,55	522,82	5 877
3200	0,7829	592,96	2 256	41 780
6400	1,6937	2 341	9 322	OoT
12800	3,7084	11 141	40 921	OoT
25600	5,5	46 897	OoT	OoT
51200	11,7	OoT	OoT	OoT

Loop1 => O(n*log(n))

Loop2 => O(n^2*log(n))

Loop3 => O(n^2*log(n))

Loop4 => O(n^3)

All the times accord to its respective expected complexity.

Activity 2: Creation of iterative models of a given time complexity

Loop5 => O(n^2log2n), **Loop6** => O(n^3log n) and **Loop7** => O(n^4).

TABLE2 (times in milliseconds WITHOUT OPTIMIZATION)

n	tLoop5	tLoop6	tLoop7

100	11,3	114	1 397
200	36,8	904	20 699
400	186,4	7 802	OoT
800	889,6	69 007	OoT
1600	4 460	OoT	OoT
3200	19 433	OoT	OoT
6400	OoT	OoT	OoT

All the times are concorded with each respect complexity.

Activity 3: Comparison of two algorithms

TABLE3 (times in milliseconds WITHOUT OPTIMIZATION)

n	t1	t2	t1/t2
100	0,016	0,41	0,03902439
200	0,033	1,54	0,02142857
400	0,0701	7,63	0,00918742
800	0,166	33,87	0,00490109
1600	0,3674	129,55	0,00283597
3200	0,7829	592,96	0,00132033
6400	1,6937	2 341	0,00072349
12800	3,7084	11 141	0,00033286
25600	5,5	46 897	0,00017278
51200	11,7	OoT	-

It can be seen that t1/t2 tends to 0, so as it is lower than 1, t2 has a higher complexity.

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Activity 4: Comparison of two algorithms

TABLE4 (times in milliseconds WITHOUT OPTIMIZATION)

n	t3	t2	t3/t2
100	1,56	0,41	3,8048
200	6,35	1,54	4,1233
400	27,7	7,63	3,6304
800	123,85	33,87	3,6566
1600	522,82	129,55	4,0356
3200	2 256	592,96	3,8046
6400	9 322	2 341	3,9820
12800	40 921	11 141	3,6730
25600	OoT	46 897	-
51200	OoT	OoT	-

The division is higher than 1, so t3 has a higher complexity, although, we can not see the division increment and increment, it varies between 3,8 and that's its due to although they have the same type of complexity, no the exact same, being the complexity of t3 2n*log(n/2) and t2 n*log3(n/2). Indeed, 2*(log2(x)/log3(x) = give as 3,16 a similar relation to the obtained.

Activity 5: Same algorithm in different development environments.

TABLE5 (times in milliseconds)

n	tLoop4(Python)	tLoop4(Java not Optimized	tLoop4(Java optimized)	t42/t41	t43/t42

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200	2,4	4,574	0,84	0,1906	0,1836
400	10	34,8	4,97	0,3480	0,1428
800	160	268,1	30,65	0,1676	0,1143
1600	1 300	2 120	205,98	0,1631	0,0972
3200	12 300	16 724	1 396,8	0,1360	0,0835
6400	OoT	OoT	11 116	-	0,0831

It’s easy to see how there are no surprises. t42 is the slower and it makes sense as having the JVM without optimizing the code adds an extra layer to the execution, and t43 is the fastest proving once again the power of the optimization of the JVM.